

CAMBRIDGE INTERNATIONAL AS & A LEVEL CHEMISTRY (9701)

Topic 3 — Chemical Bonding

Topical Past Paper Question Pack & Comprehensive Mark Scheme

CANDIDATE	Urwah
SYLLABUS	Cambridge International AS Level Chemistry (9701)
TOPIC COVERAGE	3.1 Electronegativity · 3.2 Ionic · 3.3 Metallic · 3.4 Covalent & Dative · 3.5 VSEPR · 3.6 IMF · 3.7 Dot-and-Cross
EXAM SERIES	Past Paper Compilation (2019 – 2025 Paper 2 Theory & Paper 1)
TOTAL MARKS	226 Marks (50 Questions)

HOW TO USE THIS PACK

1. Complete each question carefully in the answer spaces provided, paying close attention to mark allocations.
2. Questions are classified into Easy and Hard tiers to guide progression from core recall to advanced data analysis.
3. Every question includes authentic Cambridge examination codes (e.g. 9701/22/M/J/22) for cross-referencing.
4. Full examiner marking points with point-by-point breakdowns are provided at the end of the document.

SYLLABUS SUBTOPICS COVERED

3.1 Electronegativity and bonding

— Pauling scale, periodic trends across periods and groups, non-polar vs polar bonds, dipole moment

3.2 Ionic bonding

— Electrostatic attraction between oppositely charged ions, giant ionic lattices, physical properties

3.3 Metallic bonding

— Lattice of positive metal ions in a sea of delocalised electrons, electrical/thermal conductivity, malleability

3.4 Covalent & coordinate bonding— Shared electron pairs, orbital overlap (sigma and pi bonds), dative covalent bonds (NH_4^+ , AlCl_3)**3.5 Shapes of molecules (VSEPR)**— Bond pair/ lone pair repulsion (2 to 6 electron pairs), bond angles (180° , 120° , 109.5° , 107° , 104.5° , 90°)**3.6 Intermolecular forces**

— London dispersion (induced dipole), permanent dipole-dipole, and hydrogen bonding (N, O, F); boiling points

3.7 Dot-and-cross diagrams

— Outer shell representations of ionic, covalent, and coordinate bonded species; octet expansion

SUBTOPIC 3.1 — ELECTRONEGATIVITY & BOND POLARITY

1 Defining Electronegativity and Pauling Scale — 9701/22/M/J/21/Q2(a)

EASY · Syllabus 3.1

Electronegativity is a fundamental property that dictates the nature of chemical bonding between atoms.

- (a) Define the term electronegativity.

[2]

- (b) Name the most electronegative element in the Periodic Table and state its value on the Pauling scale.

[1]

2 Pauling Electronegativity Trend Across Period 3 — 9701/21/O/N/22/Q2(a)–(c)

HARD · Syllabus 3.1

Fig. 2.1 displays the variation in Pauling electronegativity across the elements of Period 3 from sodium to chlorine.

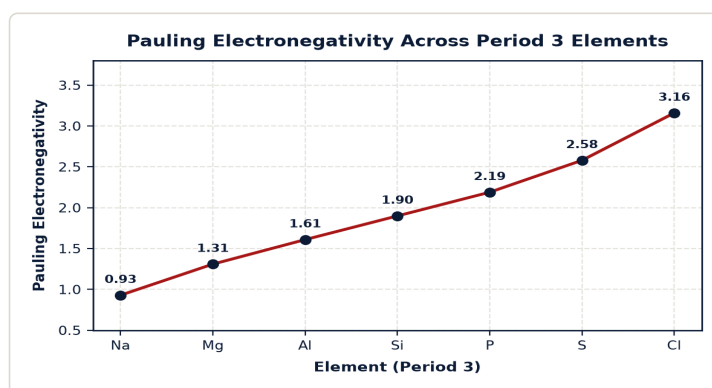


Fig. 2.1 Pauling electronegativity values across Period 3 elements

- (a) Describe the trend in electronegativity across Period 3 shown in Fig. 2.1.

[1]

- (b) Explain this trend across Period 3 in terms of nuclear charge, atomic radius, and shielding.

[3]

- (c) Explain why argon is not assigned a Pauling electronegativity value on this scale.

[1]

3 Bond Polarity and Permanent Dipoles — 9701/12/M/J/22/Q4

EASY · Syllabus 3.1

A covalent bond between two atoms of differing electronegativities is polar.

- (a) Draw a polar covalent bond between carbon and chlorine in chloromethane, CH_3Cl , using delta+ and delta- notation and a dipole moment arrow ($\rightarrow$).

[2]

- (b) Explain why tetrachloromethane, CCl_4 , contains polar bonds but possesses no overall permanent dipole moment. [2]
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4 Predicting Bond Character from Electronegativity Difference — 9701/22/F/M/23/Q1(b)

HARD · Syllabus 3.1

The difference in electronegativity ($\Delta\chi$) between bonded atoms indicates whether bonding is primarily non-polar covalent, polar covalent, or ionic.

- (a) Pauling electronegativity values are: C = 2.55, H = 2.20, Cl = 3.16, Na = 0.93. Deduce the predominant bonding character in C-H, C-Cl, and NaCl. [3]
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- (b) Explain why aluminium chloride, AlCl_3 , behaves predominantly as a covalent molecular substance at room temperature despite being a metal-nonmetal compound. [2]
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5 Electronegativity Trends Down Group 17 — 9701/11/O/N/23/Q4

EASY · Syllabus 3.1

Consider the halogens: F, Cl, Br, and I.

- (a) State and explain the trend in electronegativity down Group 17. [3]
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- (b) Arrange the following bonds in order of increasing polarity: C-F, C-Cl, C-Br, C-I. [1]
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6 Polarity in Triatomic Molecules: CO_2 vs SO_2 — 9701/22/M/J/23/Q3(b)

HARD · Syllabus 3.1

Both carbon dioxide, CO_2 , and sulfur dioxide, SO_2 , contain polar bonds between a central nonmetal atom and oxygen atoms.

- (a) State the shape and bond angle of CO_2 and explain why CO_2 is a non-polar molecule. [2]
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- (b) State the shape and approximate bond angle of SO_2 and explain why SO_2 has a permanent dipole moment. [3]
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7 Dipole Moments in Halogenoalkanes — 9701/13/M/J/21/Q4

EASY · Syllabus 3.1

Consider the isomers cis-1,2-dichloroethene and trans-1,2-dichloroethene.

- (a) Deduce which isomer possesses an overall permanent dipole moment. [1]
- A cis-1,2-dichloroethene only
B trans-1,2-dichloroethene only
C Both cis and trans isomers
D Neither isomer

- (b) Explain your choice in part (a) with reference to molecular symmetry. [2]

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SUBTOPIC 3.2 — IONIC BONDING & LATTICE PROPERTIES

8 Definition of Ionic Bonding and Giant Lattice — 9701/21/M/J/22/Q2(a)

EASY · Syllabus 3.2

Sodium chloride, NaCl, is an archetypal ionic compound.

- (a) Define ionic bonding. [2]

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- (b) Describe the structure of sodium chloride in terms of its giant ionic lattice and coordination numbers. [2]

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9 Comparing Melting Points of Ionic Compounds: NaCl vs MgO — 9701/22/O/N/22/Q2(b)

HARD · Syllabus 3.2

Sodium chloride, NaCl, has a melting point of 801 °C, whereas magnesium oxide, MgO, melts at 2852 °C.

- (a) State two factors that govern the strength of an ionic bond (lattice energy). [2]

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- (b) Explain, in terms of ionic charges and radii, why the melting point of MgO is vastly higher than that of NaCl. [3]

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10 Electrical Conductivity of Ionic Compounds — 9701/11/F/M/22/Q3

EASY · Syllabus 3.2

Pure solid sodium chloride is an electrical insulator, but becomes a conductor when molten or dissolved in water.

- (a) Explain why solid sodium chloride does not conduct electricity. [2]

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- (b) Explain why aqueous or molten sodium chloride conducts electricity. [2]

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11 Brittleness of Ionic Crystals — 9701/22/F/M/24/Q2(a)

EASY · Syllabus 3.2

When a sharp mechanical stress (such as a hammer blow) is applied to an ionic crystal, it cleaves along clean planes rather than deforming.

- (a) Explain, with the aid of a diagram or concise description, why ionic crystals are brittle. [3]

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12 Hydration and Solubility of Ionic Solids — 9701/23/M/J/23/Q2(a)

HARD · Syllabus 3.2

When an ionic solid dissolves in water, the process involves lattice breaking and ion hydration.

- (a) Describe the interaction between water molecules and a dissolved sodium cation, Na^+ . [2]

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- (b) Describe the interaction between water molecules and a dissolved chloride anion, Cl^- . [2]

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- (c) Explain why barium sulfate, BaSO_4 , is virtually insoluble in water despite being an ionic compound. [2]

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13 Polarisation and Covalent Character in Ionic Compounds — 9701/21/O/N/24/Q2(a)

HARD · Syllabus 3.2

Fajans' rules govern the degree of covalent character in ionic compounds.

- (a) State two conditions regarding the cation and anion that favour high covalent character in an ionic compound. [2]

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- (b) Compare the covalent character in lithium iodide, LiI, and potassium fluoride, KF. Justify your comparison fully. [3]

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14 Thermal Stability of Group 2 Carbonates and Nitrates — 9701/22/M/J/24/Q3(a)

HARD · Syllabus 3.2

The thermal decomposition temperature of Group 2 carbonates increases down the group from MgCO_3 to BaCO_3 .

- (a) Explain this trend in thermal stability in terms of the charge density and polarizing power of the Group 2 cations. [4]

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SUBTOPIC 3.3 — METALLIC BONDING & PROPERTIES

15 Model of Metallic Bonding — 9701/12/O/N/21/Q3

EASY · Syllabus 3.3

Metals possess unique physical properties attributed to their lattice structure.

- (a) Describe metallic bonding. [2]

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- (b) Explain why metals are good conductors of electricity in both solid and liquid states. [2]

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16 Melting Point Trends Across Period 3 Metals — 9701/22/M/J/22/Q3(a)

HARD · Syllabus 3.3

The melting points of the metallic elements in Period 3 are: sodium (98°C), magnesium (650°C), and aluminium (660°C).

- (a) Explain why the melting point of magnesium is significantly higher than that of sodium. [3]

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- (b) Predict how the electrical conductivity of aluminium compares to that of sodium at room temperature. Justify your answer. [2]

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17 Malleability and Ductility of Metals — 9701/21/F/M/23/Q1(c)

EASY · Syllabus 3.3

Metals can be hammered into thin sheets (malleable) and drawn into wires (ductile) without fracturing.

- (a) Explain why metals are malleable, contrasting their behaviour with that of ionic crystals. [3]

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- (b) Explain why alloys (such as bronze or steel) are harder and less malleable than pure metals. [2]

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18 Thermal Conductivity in Metals — 9701/11/M/J/23/Q3

EASY · Syllabus 3.3

Copper is widely used in electrical wiring and cooking utensils.

- (a) State the two mechanisms by which heat energy is transported through a metal. [2]

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- (b) State which of the two mechanisms in part (a) is the primary contributor to the high thermal conductivity of copper. [1]

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19 Trends in Metallic Bonding Down Group 1 — 9701/22/O/N/23/Q2(b)

HARD · Syllabus 3.3

The melting points of Group 1 alkali metals decrease monotonically down the group: Li (181 °C), Na (98 °C), K (63 °C), Rb (39 °C), Cs (29 °C).

- (a) Explain why the strength of metallic bonding decreases down Group 1. [3]

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20 Comparing Giant Structures: Metallic vs Ionic vs Giant Covalent — 9701/21/M/J/24/Q2(c)

HARD · Syllabus 3.3

Table 20.1 shows properties of three unknown solids X, Y, and Z.

- Solid X: MP 1414 °C, conducts electricity in both solid and liquid states.
- Solid Y: MP 801 °C, non-conductor when solid, conducts when molten.
- Solid Z: MP 3550 °C, non-conductor in both solid and liquid states.

- (a) Deduce the type of bonding and structure present in Solid X, Solid Y, and Solid Z. [3]

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- (b) Suggest an identity for Solid Z.

[1]

SUBTOPIC 3.4 — COVALENT & COORDINATE (DATIVE) BONDING

21 Coordinate Bonding in the Aluminium Chloride Dimer — 9701/22/M/J/21/Q3(a)–(c)

HARD · Syllabus 3.4

Fig. 21.1 shows the bonding in the dimer of aluminium chloride, Al_2Cl_6 , formed in the vapour phase at moderate temperatures.

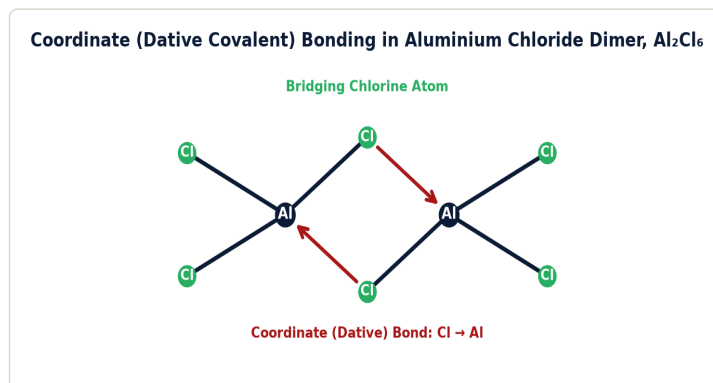


Fig. 21.1 Coordinate (dative covalent) bonding in Al_2Cl_6 dimer

- (a) Define the term coordinate (dative covalent) bond.

[2]

- (b) Using Fig. 21.1, identify which atoms donate the lone pairs and which atoms act as lone-pair acceptors.

[2]

- (c) State the total number of covalent bonds and the number of dative bonds present in one Al_2Cl_6 molecule.

[2]

22 Formation and Structure of the Ammonium Ion — 9701/21/O/N/21/Q2(b)

EASY · Syllabus 3.4

Ammonia reacts with hydrogen chloride gas to form solid ammonium chloride: $\text{NH}_3(\text{g}) + \text{HCl}(\text{g}) \rightarrow \text{NH}_4\text{Cl}(\text{s})$.

- (a) Describe how the dative covalent bond in the ammonium ion, NH_4^+ , is formed from an ammonia molecule and a hydrogen ion.

[2]

- (b) Explain why all four N–H bonds in the ammonium ion have identical bond lengths and bond energies once the ion has formed.

[2]

23 Sigma (sigma) and Pi (pi) Bonds in Hydrocarbons — 9701/22/F/M/22/Q3(a)

HARD · Syllabus 3.4

Covalent bonds can be classified as sigma (sigma) or pi (pi) bonds based on orbital overlap geometry.

- (a)** Describe the difference between a sigma bond and a pi bond in terms of how atomic orbitals overlap. **[2]**

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- (b)** State the number of sigma bonds and pi bonds present in one molecule of ethene, C₂H₄. **[2]**

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- (c)** State the number of sigma bonds and pi bonds present in one molecule of propyne, CH₃-CCH. **[2]**

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24 Bond Energy and Bond Length Correlations — 9701/12/M/J/23/Q4

EASY · Syllabus 3.4

Table 24.1 lists carbon-carbon bond lengths and average bond energies:

- C-C single bond: length 0.154 nm, energy 347 kJ mol⁻¹
- C=C double bond: length 0.134 nm, energy 612 kJ mol⁻¹
- CC triple bond: length 0.120 nm, energy 838 kJ mol⁻¹

- (a)** Describe and explain the relationship between bond order, bond length, and bond energy. **[3]**

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- (b)** Explain why the bond energy of a C=C double bond (612 kJ mol⁻¹) is less than twice the bond energy of a C-C single bond (347 kJ mol⁻¹). **[2]**

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25 Dative Bonding in Carbon Monoxide, CO — 9701/21/M/J/23/Q3(a)

HARD · Syllabus 3.4

Carbon monoxide, CO, possesses a triple bond between carbon and oxygen.

- (a)** Describe the composition of the triple bond in carbon monoxide in terms of covalent and dative covalent bonds. **[2]**

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- (b)** Explain why carbon monoxide can act as a ligand in transition metal complexes, such as haemoglobin. **[2]**

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26 Hydronium Ion, H_3O^+ , Formation — 9701/11/O/N/22/Q4

EASY · Syllabus 3.4

When acids dissolve in aqueous solution, protons hydrate to form oxonium (hydronium) ions, H_3O^+ .

- (a) Draw a displayed formula of the H_3O^+ ion, indicating the dative covalent bond with an arrow ($\rightarrow$). [2]

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- (b) Predict the shape and bond angle of the H_3O^+ ion using VSEPR theory. [2]

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27 Hybridisation: sp , sp^2 , and sp^3 — 9701/22/O/N/24/Q2(a)

HARD · Syllabus 3.4

Carbon atoms undergo orbital hybridisation to optimize bonding geometries.

- (a) State the hybridisation of each carbon atom in: ethane (C_2H_6), ethene (C_2H_4), and ethyne (C_2H_2). [3]

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- (b) Describe how one 2s orbital and two 2p orbitals hybridise to form sp^2 hybrid orbitals. [2]

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28 Electron-Deficient Covalent Compounds: BF_3 — 9701/21/M/J/25/Q2(b)

EASY · Syllabus 3.4

Boron trifluoride, BF_3 , reacts readily with ammonia, NH_3 , to form an addition compound (adduct) $\text{F}_3\text{B:NH}_3$.

- (a) Explain why BF_3 is described as an electron-deficient molecule. [1]

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- (b) Describe the bond that forms between boron and nitrogen in the $\text{F}_3\text{B:NH}_3$ adduct. [2]

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- (c) State the change in the F-B-F bond angle when BF_3 forms the adduct. [2]

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SUBTOPIC 3.5 — VSEPR THEORY & MOLECULAR GEOMETRY

29 Summary of VSEPR Molecular Shapes — 9701/22/M/J/22/Q3(b)–(c)

EASY · Syllabus 3.5

Fig. 29.1 summarizes key molecular geometries predicted by Valence Shell Electron Pair Repulsion (VSEPR) theory.

Summary of VSEPR Molecular Shapes, Angles and Examples				
Linear	Trigonal Planar	Tetrahedral	Pyramidal	Non-linear / Bent
180°	120°	109.5°	107°	104.5°
BeCl ₂ , CO ₂	BF ₃ , AlCl ₃	CH ₄ , NH ₄ ⁺	NH ₃ , PCl ₃	H ₂ O, H ₂ S

Fig. 29.1 Fundamental VSEPR molecular shapes, bond angles, and representative molecules

- (a) State the fundamental postulate of VSEPR theory. [2]
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- (b) Using Fig. 29.1, state the shape and bond angle of methane (CH₄) and water (H₂O). [2]
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- (c) Explain why the bond angle in water (104.5°) is smaller than that in methane (109.5°), despite both having four electron pairs around the central atom. [3]
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30 Expanded Octet Geometries: PCl₅ and SF₆ — 9701/21/O/N/23/Q2(b)

HARD · Syllabus 3.5

Period 3 elements can expand their octet by utilizing vacant 3d orbitals in bonding.

- (a) Deduce the number of bond pairs and lone pairs around phosphorus in phosphorus pentachloride, PCl₅. State its molecular shape and bond angles. [3]
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- (b) Deduce the molecular shape and all bond angles of sulfur hexafluoride, SF₆. [2]
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31 Shape of Xenon Tetrafluoride, XeF₄ — 9701/22/F/M/24/Q3(a)

HARD · Syllabus 3.5

Xenon forms the noble-gas compound xenon tetrafluoride, XeF₄.

- (a) Determine the number of bonding pairs and lone pairs of electrons surrounding the central xenon atom in XeF₄. [2]

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- (b) Deduce the molecular shape of XeF₄ and state the F–Xe–F bond angle. [2]

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- (c) Explain why the lone pairs adopt axial positions opposite one another. [1]

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32 Shape of Chlorine Trifluoride, ClF₃ — 9701/23/M/J/22/Q2(b)

HARD · Syllabus 3.5

Chlorine trifluoride is an extremely reactive interhalogen compound.

- (a) State the total number of valence electrons around the central chlorine atom in ClF₃, and deduce the number of bond pairs and lone pairs. [2]

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- (b) Deduce the molecular geometry (shape) and approximate bond angles in ClF₃. [2]

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33 Shapes of Polyatomic Inorganic Ions: CO₃²⁻, NO₃⁻, SO₄²⁻ — 9701/11/M/J/24/Q3

EASY · Syllabus 3.5

The shapes of oxyanions can be determined using VSEPR theory.

- (a) State the shape and bond angle of a carbonate ion, CO₃²⁻. [2]

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- (b) State the shape and bond angle of a sulfate ion, SO₄²⁻. [2]

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34 Shape of Phosphorus Trichloride, PCl₃ — 9701/12/O/N/23/Q3

EASY · Syllabus 3.5

Phosphorus trichloride, PCl₃, is a volatile liquid.

- (a) What is the shape and bond angle of PCl_3 ? [1]
A Trigonal planar, 120°
B Trigonal pyramidal, 107°
C Tetrahedral, 109.5°
D T-shaped, 90°

- (b) Explain why the bond angle in PCl_3 is slightly less than 109.5° . [2]

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35 Bond Angles in Organic Molecules — 9701/22/M/J/23/Q4(a)

HARD · Syllabus 3.5

Consider the molecule prop-2-en-1-ol: $\text{CH}_2=\text{CH}-\text{CH}_2-\text{OH}$.

- (a) State the bond angle around the $\text{C}(1)$ atom (in $=\text{CH}_2$). [1]

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- (b) State the bond angle around the $\text{C}(3)$ atom (in $-\text{CH}_2-$). [1]

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- (c) State the $\text{C}-\text{O}-\text{H}$ bond angle around the oxygen atom and justify your value. [2]

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36 Deducing Unknown Geometry from Electron Pair Counting — 9701/21/O/N/24/Q3(b)

HARD · Syllabus 3.5

Consider the interhalogen cation ICl_2^+ and the anion ICl_2^- .

- (a) Determine the number of bond pairs and lone pairs on the central iodine atom in ICl_2^+ , and predict its shape. [3]

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- (b) Determine the number of bond pairs and lone pairs on the central iodine atom in ICl_2^- , and predict its shape. [3]

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SUBTOPIC 3.6 — INTERMOLECULAR FORCES & PHYSICAL PROPERTIES

37 Boiling Point Anomalies of Binary Hydrides — 9701/22/M/J/21/Q4(a)-(d)

HARD · Syllabus 3.6

Fig. 37.1 shows the boiling points of the binary hydrides of the elements in Groups 14, 15, 16, and 17 across Periods 2 to 5.

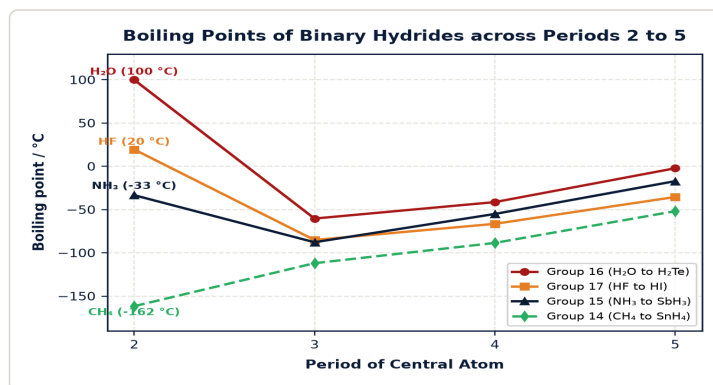


Fig. 37.1 Boiling points of binary hydrides of Groups 14, 15, 16, and 17

- (a) Explain why the boiling points increase steadily down Group 14 from CH₄ to SnH₄. [3]

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- (b) Explain why H₂O, HF, and NH₃ have anomalously high boiling points compared to the hydrides of heavier elements in their respective groups. [3]

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- (c) Explain why the boiling point of water (100 °C) is significantly higher than that of hydrogen fluoride (20 °C), even though fluorine is more electronegative than oxygen. [3]

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38 Three Types of Intermolecular Forces — 9701/21/O/N/21/Q3(a)

EASY · Syllabus 3.6

Intermolecular forces dictate the boiling points, solubilities, and physical states of molecular substances.

- (a) Name the three types of intermolecular forces in order of generally increasing typical strength. [2]

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- (b) Describe how instantaneous dipole-induced dipole (London dispersion) forces originate. [3]

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39 Hydrogen Bonding Requirements — 9701/22/F/M/22/Q3(b)

EASY · Syllabus 3.6

Hydrogen bonding is a special, strong category of permanent dipole-dipole attraction.

- (a) State the two essential structural conditions required for hydrogen bonding to occur between molecules. [2]

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- (b) Draw a diagram showing a hydrogen bond between two ethanol ($\text{C}_2\text{H}_5\text{OH}$) molecules. Include all lone pairs and partial charges (δ^+ and δ^-). [3]

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40 Density Anomaly of Ice Compared to Liquid Water — 9701/11/F/M/23/Q3

HARD · Syllabus 3.6

Unlike almost all other substances, solid ice floats on liquid water at 0°C .

- (a) Explain why solid ice has a lower density than liquid water at 0°C . [3]

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- (b) State one biological consequence of the fact that ice floats on water. [1]

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41 Comparing Boiling Points of Halogens — 9701/12/M/J/24/Q4

EASY · Syllabus 3.6

The boiling points of the halogens are: F_2 (-188°C), Cl_2 (-34°C), Br_2 (59°C), I_2 (184°C).

- (a) Identify the only type of intermolecular force present in pure liquid halogens. [1]

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- (b) Explain why the boiling point increases steadily from fluorine to iodine. [2]

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42 Branched vs Straight-Chain Alkane Boiling Points — 9701/22/O/N/23/Q4(a)

HARD · Syllabus 3.6

Pentane ($\text{CH}_3(\text{CH}_2)_3\text{CH}_3$) has a boiling point of 36°C , whereas its branched isomer 2,2-dimethylpropane ($\text{C}(\text{CH}_3)_4$) boils at 9.5°C , despite both having the molecular formula C_5H_{12} .

- (a) Explain why straight-chain pentane has a higher boiling point than branched 2,2-dimethylpropane. [3]

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43 Solubility in Polar vs Non-Polar Solvents — 9701/21/M/J/22/Q3(c)

EASY · Syllabus 3.6

Iodine, I₂, is virtually insoluble in water but readily dissolves in hexane, C₆H₁₄.

- (a) Explain why iodine is poorly soluble in water. [2]

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- (b) Explain why iodine dissolves readily in hexane. [2]

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44 Boiling Point Comparison: Propan-1-ol vs Propanone vs Butane — 9701/22/F/M/25/Q3(a)

HARD · Syllabus 3.6

Table 44.1 lists three organic compounds with similar molecular masses:

- Butane (Mr = 58): BP -0.5 °C
- Propanone (Mr = 58): BP 56 °C
- Propan-1-ol (Mr = 60): BP 97 °C

- (a) Identify the strongest type of intermolecular force present in each pure compound. [3]

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- (b) Explain the order of boiling points: Butane < Propanone < Propan-1-ol. [3]

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45 Viscosity and Surface Tension Trends in Alcohols — 9701/11/O/N/24/Q4

EASY · Syllabus 3.6

Ethanol (CH₃CH₂OH) is a free-flowing liquid, whereas glycerol (propane-1,2,3-triol, CH₂OH-CHOH-CH₂OH) is a viscous, syrupy liquid.

- (a) Explain why glycerol has a vastly higher viscosity and boiling point than ethanol. [3]

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SUBTOPIC 3.7 — DOT-AND-CROSS REPRESENTATIONS

46 Dot-and-Cross Diagram of Methanoic Acid, HCO_2H — 9701/21/M/J/21/Q4(d)

EASY · Syllabus 3.7

Methanoic acid has the molecular formula HCO_2H . The central carbon atom is bonded to a hydrogen atom, a carbonyl oxygen atom, and a hydroxyl group.

- (a) Draw a 'dot-and-cross' diagram, showing outer-shell electrons only, to represent the bonding in methanoic acid, HCO_2H .

[2]

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- (b) State the total number of non-bonding lone pairs of electrons in a molecule of methanoic acid.

[1]

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47 Dot-and-Cross Diagram for Nitrogen Gas, N_2 , and Cyanide Ion, CN^- — 9701/22/O/N/21/Q4(a)

HARD · Syllabus 3.7

Both dinitrogen (N_2) and the cyanide ion (CN^-) contain a triple bond between two atoms.

- (a) Draw a 'dot-and-cross' diagram of a nitrogen molecule, N_2 , showing outer electrons only.

[2]

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- (b) Draw a 'dot-and-cross' diagram of a cyanide anion, CN^- , indicating the origin of the negative charge.

[2]

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48 Dot-and-Cross Diagram with Expanded Octet: SF_6 and PCl_5 — 9701/23/M/J/22/Q3(c)

HARD · Syllabus 3.7

Sulfur hexafluoride, SF_6 , contains 12 electrons in the valence shell of sulfur.

- (a) Draw a 'dot-and-cross' diagram for SF_6 , showing all valence electrons around sulfur and fluorine atoms.

[3]

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- (b) Explain why sulfur can expand its octet to form SF_6 , whereas oxygen in the same group can only form OF_2 and cannot form OF_6 .

[2]

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Dot-and-Cross Diagram for Dative Bonded Species: NH_4^+ and H_3O^+ — 9701/12/M/J/23/Q3

EASY · Syllabus 3.7

Dative covalent bonds can be represented in dot-and-cross diagrams where both shared electrons originate from the same atom. [2]

- (a) Draw a 'dot-and-cross' diagram for an ammonium ion, NH_4^+ , using dots for nitrogen electrons, crosses for three hydrogen electrons, and an open circle for the dative electron pair.

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- (b) Draw a 'dot-and-cross' diagram for the hydronium ion, H_3O^+ . [2]

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Synoptic Problem: Chemical Bonding, Geometry & Intermolecular Forces — 9701/22/O/N/25/Q3

HARD · Syllabus 3.7

Hydrazine, N_2H_4 , is a rocket propellant that is miscible with water in all proportions.

- Boiling point of hydrazine: 114°C
- Melting point of hydrazine: 2°C

- (a) Draw a 'dot-and-cross' diagram for a hydrazine molecule, N_2H_4 , showing outer electrons only. [2]

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- (b) Deduce the shape and bond angle around each nitrogen atom in hydrazine. [2]

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- (c) Explain why hydrazine has a boiling point higher than water (114°C vs 100°C) despite having a lower electronegativity difference than H_2O . [3]

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Mark Scheme — Topic 3 — Chemical Bonding

CAMBRIDGE INTERNATIONAL AS LEVEL CHEMISTRY (9701) — THEORY MARKING POINTS

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
1(a)	The ability / power of an atom to attract the shared pair of electrons (1); in a covalent bond (1)	2
1(b)	Fluorine / F, with a value of 4.0 (1)	1
2(a)	Electronegativity increases steadily across Period 3 from Na to Cl (1)	1
2(b)	Nuclear charge increases across the period (+11 to +17) (1); Atomic radius decreases / bonding electrons are closer to the nucleus (1); Shielding remains similar (electrons added to $n=3$), resulting in a greater effective nuclear attraction for the shared pair (1)	3
2(c)	Argon has a stable complete octet and does not form standard covalent bonds (1)	1
3(a)	C(δ^+) - Cl(δ^-) with correct charges (1); Dipole arrow pointing from C towards Cl with cross at C ($\rightarrow$) (1)	2
3(b)	CCl ₄ has a regular symmetrical tetrahedral geometry (1); The four individual C-Cl bond dipoles cancel each other out completely (vector sum = 0) (1)	2
4(a)	C-H: non-polar covalent ($\delta = 0.35$) (1); C-Cl: polar covalent ($\delta = 0.61$) (1); NaCl: ionic ($\delta = 2.23$) (1)	3
4(b)	Al ³⁺ ion has a high charge (+3) and a very small ionic radius, giving it an extremely high charge density (1); It polarizes the electron cloud of the large chloride ion (Cl ⁻) to such an extent that electron sharing (covalency) occurs (1)	2
5(a)	Electronegativity decreases down Group 17 (1); Atomic radius increases and there are more inner shielding electron shells (1); Outermost bonding pair is further from the nucleus and experiences weaker attraction (1)	3
5(b)	C-I < C-Br < C-Cl < C-F (1)	1
6(a)	Linear, 180° (1); The two equal C=O bond dipoles point in opposite directions and cancel out completely (1)	2
6(b)	Non-linear / bent, approximately 117° to 119° (1); Central S atom has a lone pair of electrons creating an asymmetrical charge distribution (1); The bond dipoles do not cancel, giving a net permanent dipole towards oxygen (1)	3
7(a)	A — cis-1,2-dichloroethene only (1)	1
7(b)	In the trans isomer, the two C-Cl bond dipoles point in directly opposite directions across the double bond and cancel out completely (1); In the cis isomer, both C-Cl dipoles point to the same side, creating a net molecular dipole (1)	2
8(a)	The electrostatic attraction (1); between oppositely charged ions (1)	2
8(b)	Giant three-dimensional cubic lattice of alternating Na ⁺ and Cl ⁻ ions (1); Each ion has a coordination number of 6 (surrounded octahedrally by 6 oppositely charged ions) (1)	2
9(a)	Charges on the ions (1); Radii of the ions / sum of ionic radii (1)	2
9(b)	Mg ²⁺ and O ²⁻ ions carry 2+ and 2- charges whereas Na ⁺ and Cl ⁻ carry 1+ and 1- charges (1); Mg ²⁺ and O ²⁻ ions have smaller radii than Na ⁺ and Cl ⁻ (1); The electrostatic attraction between the 2+/2- ions is roughly 4 times stronger, requiring far more thermal energy to break the lattice (1)	3
10(a)	In the solid lattice, ions are held rigidly in fixed positions by strong electrostatic forces (1); There are no mobile ions or delocalised electrons to carry charge (1)	2
10(b)	When molten or dissolved in water, the giant ionic lattice breaks down (1); The ions become free to move and carry electric current towards electrodes (1)	2
11(a)	Mechanical stress / force causes layers of ions to shift relative to one another (1); Ions of like charge are forced into adjacent positions (+ next to +, - next to -) (1); Strong electrostatic repulsion between like-charged ions causes the crystal lattice to fracture/shatter (1)	3
12(a)	Ion-dipole attraction (1); The partially negative oxygen atoms (δ^-) of polar water molecules are attracted to and surround the positive Na ⁺ ion (1)	2
12(b)	Ion-dipole attraction / hydrogen bonding (1); The partially positive hydrogen atoms (δ^+) of water molecules are attracted to and surround the negative Cl ⁻ ion (1)	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
12(c)	The lattice energy of BaSO ₄ is extremely exothermic due to 2+ and 2- charges (1); The energy released by hydration of Ba ²⁺ and SO ₄ ²⁻ ions is insufficient to overcome the strong lattice enthalpy (1)	2
13(a)	Small cation with high charge (high polarizing power) (1); Large anion with high charge (high polarizability) (1)	2
13(b)	LiI has much greater covalent character than KF (1); Li ⁺ is very small with high polarizing power, and I ⁻ is a very large anion whose electron cloud is easily distorted (1); K ⁺ has low polarizing power and F ⁻ is small and tightly held, resulting in essentially purely ionic bonding in KF (1)	3
14(a)	All Group 2 cations carry a 2+ charge, but ionic radius increases down the group from Mg ²⁺ to Ba ²⁺ (1); Charge density of the cation decreases down the group (1); Smaller cations (Mg ²⁺) have higher polarizing power and distort the large, polarizable carbonate (CO ₃ ²⁻) electron cloud more strongly (1); This weakens the C-O bond within the carbonate ion, lowering the activation energy for decomposition (1)	4
15(a)	Electrostatic attraction (1); between a regular lattice of positive metal ions (cations) and a sea of delocalised electrons (1)	2
15(b)	Delocalised valence electrons are free to move throughout the entire metallic lattice (1); Under an applied potential difference, they drift towards the positive terminal, conducting electrical current (1)	2
16(a)	Mg ²⁺ carries a 2+ charge whereas Na ⁺ carries a 1+ charge (1); Mg contributes 2 delocalised electrons per atom compared to 1 for Na (greater electron density) (1); Mg ²⁺ has a smaller ionic radius than Na ⁺ , creating a vastly stronger electrostatic attraction to delocalised electrons (1)	3
16(b)	Aluminium has higher electrical conductivity than sodium (1); Each Al atom contributes 3 delocalised electrons to the conduction band compared to 1 for Na, providing a higher carrier concentration (1)	2
17(a)	In a metal, layers of positive ions can slide over one another when stress is applied (1); The non-directional sea of delocalised electrons maintains electrostatic bonding across all directions without repulsion (1); Unlike ionic solids where like charges align and repel (1)	3
17(b)	Alloy atoms have different atomic sizes compared to host metal atoms (1); This disrupts the regular layered lattice structure, preventing layers from sliding easily over one another (1)	2
18(a)	Movement / kinetic transport of energetic delocalised electrons (1); Lattice vibrations (phonons) transmitted through closely packed cations (1)	2
18(b)	Kinetic transport by delocalised electrons (1)	1
19(a)	All Group 1 cations have a 1+ charge and contribute 1 delocalised electron per atom (1); The ionic radius of the cation increases down the group from Li ⁺ to Cs ⁺ (1); The larger distance between the delocalised electron sea and the cation nucleus reduces electrostatic attraction (1)	3
20(a)	Solid X: metallic bonding / giant metallic lattice (1); Solid Y: ionic bonding / giant ionic lattice (1); Solid Z: covalent bonding / giant covalent (macromolecular) lattice (1)	3
20(b)	Diamond (or silicon / silicon dioxide) (1)	1
21(a)	A covalent bond in which both shared electrons are provided by the same atom (1); into an empty orbital of another atom (1)	2
21(b)	Bridging chlorine atoms donate a lone pair (1); Aluminium atoms accept the lone pair into an empty 3p orbital (1)	2
21(c)	4 standard covalent bonds (terminal Al-Cl) and 2 dative covalent bonds (or 6 total bonds with 2 dative) (2)	2
22(a)	The nitrogen atom in NH ₃ donates its lone pair of electrons (1); into the vacant 1s orbital of a hydrogen ion (H ⁺) (1)	2
22(b)	Once formed, a dative covalent bond is identical in character to a standard covalent bond (1); The positive charge and electron density are distributed symmetrically across all four N-H bonds in the tetrahedral ion (1)	2
23(a)	Sigma bond: direct 'head-on' / end-to-end overlap of orbitals along the internuclear axis (1); Pi bond: 'sideways' parallel overlap of unhybridised p orbitals above and below the internuclear axis (1)	2
23(b)	5 sigma bonds (four C-H, one C-C) and 1 pi bond (in C=C) (2)	2
23(c)	6 sigma bonds (four C-H, two C-C) and 2 pi bonds (in CC) (2)	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
24(a)	As bond order increases from single to triple, bond length decreases and bond energy increases (1); More shared electron pairs increase electrostatic attraction to nuclei (1); Nuclei are pulled closer together, requiring greater energy to break (1)	3
24(b)	A double bond consists of one sigma bond and one pi bond (1); Sideways overlap in a pi bond is less effective than head-on overlap in a sigma bond, making the pi bond weaker ($612 - 347 = 265 \text{ kJ mol}^{-1}$ for pi bond) (1)	2
25(a)	Two standard covalent bonds (one electron from C, one from O) (1); One dative covalent bond where oxygen donates a lone pair into an empty carbon orbital (1)	2
25(b)	Carbon has a lone pair of electrons (1); It donates this lone pair to form a coordinate bond with Fe^{2+} in haemoglobin (1)	2
26(a)	Oxygen bonded to three H atoms with brackets and positive charge (1); One bond drawn with an arrow pointing from O to H (1)	2
26(b)	Trigonal pyramidal (1); Bond angle approximately 107° (similar to NH_3 due to 3 bond pairs and 1 lone pair) (1)	2
27(a)	Ethane: sp^3 (1); Ethene: sp^2 (1); Ethyne: sp (1)	3
27(b)	Mixing / linear combination of one 2s and two 2p atomic orbitals (1); Producing three equivalent degenerate sp^2 hybrid orbitals arranged in a trigonal planar geometry at 120° angles (leaving one unhybridised 2p orbital) (1)	2
28(a)	The central boron atom has only 6 valence electrons (3 shared pairs), lacking a stable octet (1)	1
28(b)	Coordinate (dative covalent) bond (1); Nitrogen donates its lone pair of electrons into the empty 2p orbital of boron (1)	2
28(c)	Decreases from 120° (trigonal planar in BF_3) (1); to approximately 109.5° (tetrahedral geometry around boron in adduct) (1)	2
29(a)	Electron pairs around a central atom repel each other (1); They arrange themselves in three-dimensional space to minimize mutual repulsion (1)	2
29(b)	Methane: tetrahedral, 109.5° (1); Water: non-linear / bent, 104.5° (1)	2
29(c)	Order of repulsion: lone pair-lone pair > lone pair-bond pair > bond pair-bond pair (1); CH_4 has 4 bond pairs; H_2O has 2 bond pairs and 2 lone pairs (1); The greater repulsion from the two lone pairs in water pushes the bond pairs closer together, reducing the angle by $\sim 2.5^\circ$ per lone pair (1)	3
30(a)	5 bond pairs, 0 lone pairs (1); Trigonal bipyramidal (1); Equatorial angles 120° and axial-equatorial angles 90° (and 180° axial-axial) (1)	3
30(b)	Octahedral (6 bond pairs, 0 lone pairs) (1); All bond angles are 90° (and 180°) (1)	2
31(a)	Xe has 8 valence electrons + 4 from F = 12 electrons $\Rightarrow$ 6 pairs (1); 4 bonding pairs and 2 lone pairs (1)	2
31(b)	Square planar (1); Bond angle 90° (1)	2
31(c)	Placing the two lone pairs at 180° to each other minimizes lone pair-lone pair repulsion (1)	1
32(a)	Cl has $7 + 3 = 10$ electrons (5 pairs) (1); 3 bonding pairs and 2 lone pairs (1)	2
32(b)	T-shaped (1); Bond angle approximately 87.5° to 89° (less than 90° due to lone pair repulsion) (1)	2
33(a)	Trigonal planar (1); Bond angle 120° (1)	2
33(b)	Tetrahedral (1); Bond angle 109.5° (1)	2
34(a)	B — Trigonal pyramidal, 107° (1)	1
34(b)	Central phosphorus atom has 3 bond pairs and 1 lone pair (1); Lone pair-bond pair repulsion is stronger than bond pair-bond pair repulsion, compressing the bond angle (1)	2
35(a)	120° (sp^2 hybridised carbon, 3 regions of electron density, trigonal planar) (1)	1
35(b)	109.5° (sp^3 hybridised carbon, tetrahedral arrangement of 4 bond pairs) (1)	1
35(c)	Approximately 104.5° (or $104^\circ - 105^\circ$) (1); Oxygen has 2 bonding pairs and 2 lone pairs, identical to water (1)	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
36(a)	I has $7 - 1 = 6$ valence electrons + 2 Cl = 8 electrons (4 pairs) (1); 2 bond pairs and 2 lone pairs (1); Non-linear / bent (approx. 104.5°) (1)	3
36(b)	I has $7 + 1 = 8$ valence electrons + 2 Cl = 10 electrons (5 pairs) (1); 2 bond pairs and 3 lone pairs (1); Linear (the 3 lone pairs occupy equatorial positions at 120° , bond angle 180°) (1)	3
37(a)	Group 14 hydrides are non-polar molecules held solely by London dispersion (instantaneous dipole-induced dipole) forces (1); Down the group, the number of electrons per molecule increases (16 to 66) (1); The larger electron cloud is more polarizable, creating stronger London dispersion forces (1)	3
37(b)	O, F, and N are small, highly electronegative atoms with lone pairs (1); They form intermolecular hydrogen bonds between molecules (1); Hydrogen bonds are much stronger than London dispersion forces or permanent dipole-dipole attractions, requiring more thermal energy to break (1)	3
37(c)	Each H ₂ O molecule has 2 hydrogen atoms and 2 lone pairs on oxygen, forming an average of 2 hydrogen bonds per molecule (3D network) (1); HF has only 1 hydrogen atom per molecule, forming on average only 1 hydrogen bond per molecule (1); Water forms twice as many hydrogen bonds per molecule as HF, so total intermolecular energy is greater (1)	3
38(a)	London dispersion forces < Permanent dipole-dipole attractions < Hydrogen bonding (2 marks for all three in order, 1 mark for names without correct order) (2)	2
38(b)	Constant random motion of electrons within an atom/molecule produces an uneven temporary charge distribution (1); This creates an instantaneous dipole (1); The instantaneous dipole induces an opposite dipole in an adjacent atom/molecule, resulting in electrostatic attraction (1)	3
39(a)	A hydrogen atom bonded covalently to a small, highly electronegative atom (N, O, or F) (1); A lone pair of electrons on a neighboring N, O, or F atom (1)	2
39(b)	O(δ^-) – H(δ^+) dipole shown correctly on both molecules (1); Lone pair shown on oxygen atom (1); Hydrogen bond drawn as a dashed line from the lone pair of one oxygen to the δ^+ hydrogen of the other, with O–H...O angle linear / 180° (1)	3
40(a)	In ice, water molecules are held in a regular, open, three-dimensional tetrahedral lattice by hydrogen bonds (1); Each water molecule forms 4 hydrogen bonds, creating large spaces/cavities within the crystal structure (1); When ice melts, the rigid hydrogen-bonded network collapses and water molecules pack more closely together in the liquid state (1)	3
40(b)	Ice insulates the liquid water beneath, preventing aquatic life from freezing in lakes and ponds during winter (1)	1
41(a)	Instantaneous dipole-induced dipole forces / London dispersion forces (1)	1
41(b)	Total number of electrons per diatomic molecule increases down the group (F ₂ =18 to I ₂ =106) (1); Larger electron cloud is more polarizable, creating stronger London dispersion forces requiring more thermal energy to overcome (1)	2
42(a)	Pentane has a zig-zag chain structure with a large molecular surface area of contact between molecules (1); 2,2-dimethylpropane is spherical with a smaller surface area of contact, preventing molecules from packing closely together (1); Pentane experiences more numerous London dispersion attractions across its surface, requiring more thermal energy to separate molecules (1)	3
43(a)	Water molecules are held together by strong hydrogen bonds (1); The weak dispersion interactions between non-polar I ₂ and polar H ₂ O molecules cannot release enough energy to break the hydrogen-bonded network of water (1)	2
43(b)	Both I ₂ and hexane are non-polar molecules held by London dispersion forces (1); The solute-solvent dispersion attractions formed are similar in magnitude to the solute-solute and solvent-solvent attractions (1)	2
44(a)	Butane: London dispersion forces (1); Propanone: Permanent dipole-dipole attractions (from C=O) (1); Propan-1-ol: Hydrogen bonding (from O–H group) (1)	3
44(b)	All three have comparable electron counts and similar dispersion force magnitudes (1); Propanone has additional permanent dipole-dipole attractions which are stronger than dispersion forces, raising its BP above butane (1); Propan-1-ol has hydrogen bonding which is substantially stronger than dipole-dipole attractions, requiring the most energy to vaporize (1)	3
45(a)	Ethanol contains only one –OH group, forming limited hydrogen bonds (1); Glycerol contains three –OH groups per molecule (1); Glycerol forms extensive three-dimensional intermolecular hydrogen-bonding networks, restricting the flow of molecules and creating high viscosity (1)	3

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
46(a)	Single C-H pair, double C=O (4 shared electrons), single C-O pair, single O-H pair correctly shown (1); Two lone pairs on carbonyl oxygen, two lone pairs on hydroxyl oxygen (1)	2
46(b)	4 lone pairs (two on each oxygen atom) (1)	1
47(a)	Triple bond shown with 6 shared electrons (3 dots, 3 crosses) (1); One lone pair on each nitrogen atom (1)	2
47(b)	Triple bond shown with 6 shared electrons between C and N (1); One lone pair on N, one lone pair on C, outer bracket with 1- charge indicating the extra gained electron on carbon (1)	2
48(a)	6 shared pairs between S and F (12 electrons around S) (1); 3 lone pairs (6 non-bonding electrons) on each of the six fluorine atoms (1); Clear differentiation using dots and crosses (1)	3
48(b)	Sulfur is in Period 3 and has access to energetically accessible vacant 3d orbitals for octet expansion (1); Oxygen is in Period 2 and has only 2s and 2p orbitals (maximum capacity 8 electrons, no low-lying d orbitals) (1)	2
49(a)	Central N surrounded by 4 H atoms with brackets and + charge (1); Three standard N-H pairs (dot and cross) and one dative pair (two dots from N) (1)	2
49(b)	Central O with 1 lone pair (two dots) and 3 bonding pairs to H in brackets with + charge (1); Clear representation of dative pair from oxygen to H+ (1)	2
50(a)	Single N-N bond (2 shared electrons) and four N-H bonds (2 shared electrons each) (1); One lone pair on each nitrogen atom (1)	2
50(b)	Trigonal pyramidal (around each N) (1); Bond angle approximately 107° (3 bond pairs, 1 lone pair) (1)	2
50(c)	Hydrazine has a higher relative molecular mass ($M_r = 32.0$ vs 18.0) and more electrons (18 vs 10), resulting in significantly stronger London dispersion forces (1); Each hydrazine molecule possesses two -NH ₂ groups with two lone pairs, allowing multiple hydrogen bonds per molecule (1); The combination of stronger dispersion forces and extensive hydrogen bonding requires more energy to overcome than water's forces (1)	3